CO1 AT3 TECHNICAL PRESENTATION

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COURSE:CSA 0613 DESIGN AND ANALYSIS OF ALGORITHM

This presentation explains the analysis of the recurrence relation $T(n) = T(n-1) + n$ using the Substitution Method. It demonstrates how the recurrence is expanded step by step until the base case is reached, then simplified using the arithmetic series formula to obtain a closed-form solution. The presentation also provides a mathematical justification for each expansion step, derives the final asymptotic time complexity $\Theta(n^2)$, and interprets the algorithm's efficiency. This analysis highlights the importance of recurrence solving techniques in evaluating the performance of recursive algorithms in Design and Analysis of Algorithms (DAA).

TECHNICAL PRESENTATION

Substitution Method Analysis of the Recurrence

$T(n)=T(n-1)+n$

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Introduction

- The substitution method is a mathematical technique used to solve recurrence relations.
- It works by repeatedly expanding the recurrence until the base case is reached.
- The resulting expression is simplified using known summation formulas.
- It is commonly used to analyze the time complexity of recursive algorithms.

Problem Statement

- **Recurrence Relation**

Given:

- $T(n) = T(n-1) + n, T(1) = 1$
- **Objective:**
- Expand the recurrence step by step.
- Apply mathematical substitution.
- Derive the asymptotic time complexity.
- Evaluate the efficiency of the algorithm.

Substitution Process

- **Recurrence Expansion**

Expand recursively:

$$T(n) = T(n-1) + n$$
$$= T(n-2) + (n-1) + n$$
$$= T(n-3) + (n-2) + (n-1) + n$$

Continue until the base case:

$$T(n) = T(1) + 2 + 3 + \cdots + n$$

This transforms the recurrence into an arithmetic series.

Mathematical Derivation

Using the arithmetic series formula:

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

Substitute into the recurrence:

$$T(n) = T(1) + \frac{n(n+1)}{2} - 1$$

Ignoring constants and lower-order terms:

$$\boxed{T(n) = \Theta(n^2)}$$

Complexity Analysis

Parameter	Complexity
Time Complexity	**$\Theta(n^2)$**
Space Complexity (Recursive)	**$\Theta(n)$**
Space Complexity (Iterative)	**$\Theta(1)$**

Conclusion

- The substitution method systematically solves recurrence relations through recursive expansion.
- The recurrence $T(n)=T(n-1)+n$ reduces to the sum of the first n natural numbers.
- The closed-form solution is dominated by n^2.
- Final Time Complexity: $\Theta(n^2)$
- This analysis demonstrates how mathematical techniques are used to evaluate recursive algorithm performance in algorithm design and analysis.